

HAL Campus Evidence Desk

Private, Auditable AI for Campus Operations and Research

Global Smart Campus Technology Innovation Challenge 2026

Early-Stage EdTech Startup Stream

Applicant: HAL SUPREME / Jakob Hedrich

One-line proposition: A bounded AI evidence desk that turns fragmented campus records into cited, human-reviewable action briefs without giving the model unilateral decision authority.

Current product state

A functional MVP exists with three synthetic campus scenarios, a browser UI, JSON API, deterministic evidence validation, bounded live-model synthesis, contradiction checks, and a mandatory human-review gate.

Why this matters

Campus administrators, researchers, facilities staff, IT/security teams, and department leaders often reconcile policies, service requests, operational notes, and research requirements across disconnected systems. Manual synthesis is slow; generic AI can omit provenance or blur uncertainty. HAL Campus Evidence Desk is designed around traceability and accountable review.

Competition demo data is synthetic. No university customer, institutional pilot, student-data deployment, revenue, learning-outcome result, or compliance certification is claimed.

1. Problem, Users, and Solution

Problem

Fragmented institutional evidence creates delay and inconsistency. Existing general-purpose assistants can produce fluent answers, but a campus reviewer still needs to know: What evidence supports this statement? Were contradictory records surfaced? What is inference versus fact? Who is authorized to approve the next step?

Target users

- Campus operations and administrative staff
- Faculty and research teams
- IT and security teams
- Facilities and sustainability staff
- Department leaders preparing evidence-backed decisions

Solution

The Evidence Desk accepts a bounded case evidence set and produces:

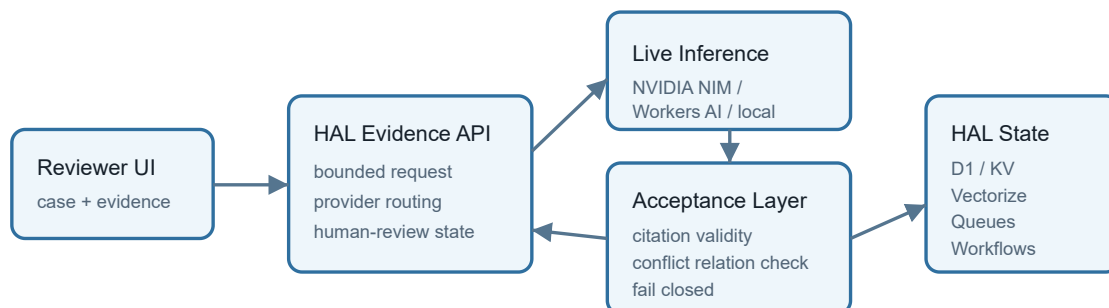
- evidence-grounded findings with explicit evidence IDs;
- contradiction surfacing without invented resolution;
- reversible advisory actions;
- explicit uncertainty;
- live-model suggestions filtered by deterministic acceptance rules; and
- a mandatory `PENDING_HUMAN_REVIEW` state.

Demonstration scenarios

1. **Conflicting policy guidance:** detects incompatible approval paths and pauses approval for human resolution.
2. **Research access request:** identifies required owner approval, purpose, retention, and secure workspace controls.
3. **Facilities energy anomaly:** prioritizes investigation of a manual air-handler override before weaker explanations.

Product boundary: admissions, grading, employment, discipline, medical decisions, and other high-impact individual decisions are intentionally outside MVP scope.

2. Technical Architecture



Core architecture principles

- **Evidence is authoritative:** model output cannot create new evidence IDs.
- **Provider independence:** model providers are replaceable; the acceptance policy is not.
- **Fail-closed controls:** invalid citations and unsupported conflict relations are rejected before display.
- **Human authority:** recommendations remain advisory and cannot self-approve.
- **Private deployment options:** sensitive evidence can remain inside institution-selected infrastructure and local/open inference paths.

Current verified infrastructure

HAL already has verified serverless inference and edge services, durable state, vector retrieval, async queues/workflows, and CI capacity. The campus product uses only the subset required for a bounded workflow.

3. Validation and Responsible AI

Current MVP validation

Measure	Current evidence
Automated tests	8/8 passing after adding false-conflict rejection coverage.
Deterministic citation validity	100% across all three canonical synthetic cases.
Unsupported material claims	0 in the deterministic canonical cases.
Seeded conflict detection	All canonical expected conflicts correctly matched.
Live provider canary	3/3 bounded NVIDIA NIM runs completed without transport/model error.
Live acceptance grounding	All accepted live items used supplied evidence IDs; accepted conflict relations matched seeded truth.
Prototype live usage	1,409 total tokens across the three validation runs; mean latency about 18.6 seconds.
Human oversight	Every report remains `PENDING_HUMAN_REVIEW`.

Responsible-AI controls

- Every accepted model item must cite supplied evidence.
- Missing or conflicting evidence is made visible.
- Unsupported conflict relationships are rejected even when the cited evidence IDs are individually valid.
- Evidence, model inference, and reviewer decisions remain distinct.
- Output size/model allowlists bound cost and variability.
- High-impact individual decisions remain out of scope.

Security and privacy direction

The demo contains no student records. Production hardening includes SSO/RBAC, least-privilege connectors, encryption, tenant isolation, configurable retention/deletion, audit logs, provider allowlists, request limits, and institution-controlled deployment boundaries.

4. Value, Business Model, and Scale-Up

Measurable value

The pilot will measure:

- citation validity and unsupported-claim rate;
- contradiction detection;
- reviewer time-to-source for each recommendation;
- time saved preparing operational briefs;
- reviewer acceptance/rejection reasons; and
- latency/cost per analysis path.

Business model hypothesis

Institution-hosted or managed deployment with a free/open public-interest core where practical, plus paid integration, managed infrastructure, support, and governance services. Pricing is not yet validated and is presented as an early-stage hypothesis rather than established revenue.

Pilot and scale-up path

1. Competition MVP and recorded demonstration.
2. One bounded institutional workflow using synthetic/public or institution-approved data.
3. Measure review time, citation quality, contradiction handling, and usability.
4. Add SSO, tenant isolation, retention, and audit controls.
5. Add institution-approved connectors and local/open provider options.
6. Expand only to workflows that pass the same provenance and human-review gates.

Implementation feasibility

The core evidence workflow already operates. A pilot-ready version is primarily authentication, integration, governance, and user-testing work. Estimated hardening time for one bounded workflow is approximately 6-8 weeks, subject to timely access to institutional integration endpoints and designated reviewers.

Why this entry is differentiated: the product is not asking a university to trust a smarter chatbot. It is offering an auditable evidence contract around AI assistance: supplied evidence in, constrained synthesis, deterministic acceptance, and accountable human review out.

Founder

Jakob Hedrich - solo founder and AI engineer behind HAL SUPREME, focused on orchestration, local/open models, auditable agent workflows, infrastructure routing, and reusable engineering systems.

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